transitions into various game stages, encompassing a game tutorial and the main experiment, which comprises multiple rounds against a range of distinct AI agents. After each round, participants are required to complete a survey. The experiment is concluded with a final survey. Comprehensive data, encompassing player trajectories and AI network parameters, are meticulously recorded and stored in the respective database. In Appendix E, screenshots from different phases of the pipeline can be found (Figs. 13-19).

The remainder of this section is devoted to introducing the Overcooked game, an overview of the Human-AI Experiment Platform, and a discussion on the design of the experimental scale.

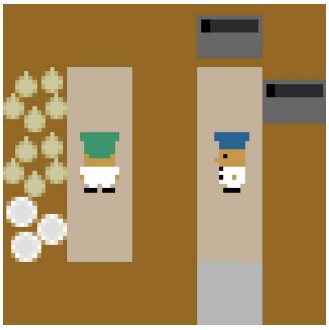
6.1 Overcooked Environment

Our paper implements the platform in the Overcooked environment (Carroll et al., 2020; Charakorn, Manoonpong, & Dilokthanakul, 2020; Knott, Carroll, Devlin, Ciosek, Hofmann, Dragan, & Shah, 2021), a simulation environment for reinforcement learning derived from the Overcooked!2 video game (Carroll et al., 2020). The Overcooked environment features a two-player collaborative game structure with shared rewards, where each player assumes the role of a chef in a kitchen, working together to prepare and serve soup for a team reward of 20 points. The environment consists of five distinct layouts: Cramped Room (Cramped Rm.), Asymmetric Advantages (Asymm. Adv.), Coordination Ring (Coord. Ring), Forced Coordination (Forced Coord.), and Counter Circuit (Counter Circ.). Visual representations of these layouts can be found in Fig. 5. The detailed introduction of five layouts is as follows.

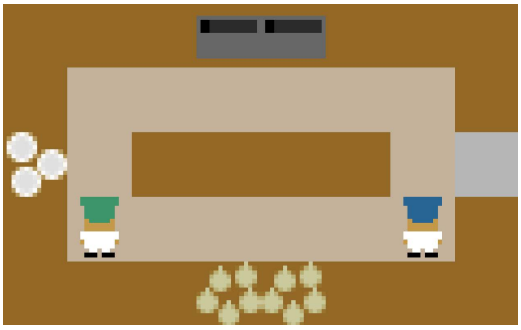
Forced Coordination. The Forced Coordination environment is designed to necessitate cooperation between the two players, as they are situated in separate, non-overlapping sections of the kitchen. Furthermore, the available equipment is distributed between these two areas, with ingredients and plates located in the left section and pots and the serving area in the right section. Consequently, the players must work together and coordinate their actions to complete a recipe and earn rewards successfully.

Counter Circuit. The Counter Circuit layout features a ring-shaped kitchen with a central, elongated table and a circular path between the table and the operational area. In this configuration, pots, onions, plates, and serving spots are positioned in four distinct directions within the operational area. Although the layout does not explicitly require cooperation, players may find themselves obstructed by narrow aisles, prompting the need for coordination to maximize rewards. One example of an advanced technique players can learn is to place onions in the middle area for quick and efficient passing, thereby enhancing overall performance.

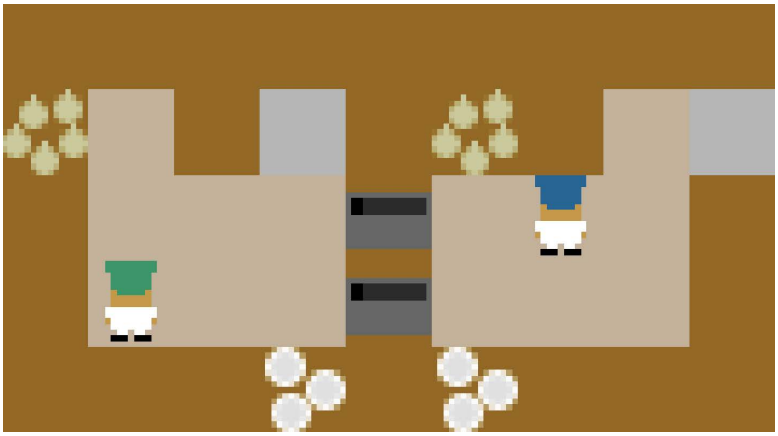
Asymmetric Advantages. In the Asymmetric Advantages layout, players are divided into two separate areas, but each player can independently complete the cooking process in their respective areas without cooperation. However, the asymmetrical arrangement of the left and right sides encourages collaboration to achieve higher rewards. Specifically, two pots are placed in the central area, accessible to both players. The areas for serving and ingredients, however, are completely distinct. The serving pot is placed near the middle on the left side and far from the middle on the right side, with the ingredients area arranged oppositely.



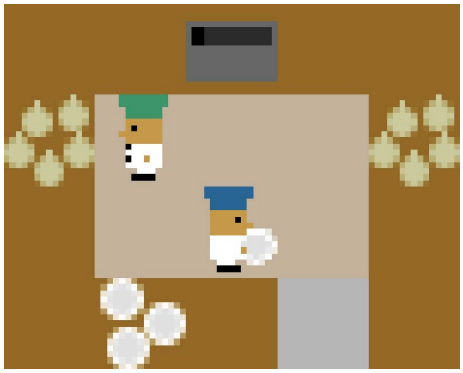
(a) Forced Coord.



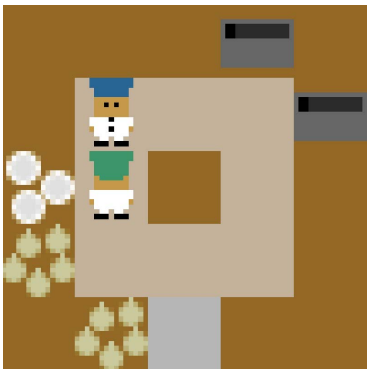
(b) Counter Circ.



(c) Asymm. Adv.



(d) Cramped Rm.



(e) Coord. Ring

Figure 5: Overcooked environment layouts. The Cramped Rm., Asymm. Adv., and Coord. Ring layouts are more conducive to higher rewards when players cooperate with different partners. On the other hand, the Forced Coord., Counter Circ., and Asymm. Adv. layouts offer distinct designs that serve as ideal testbeds to explore and foster cooperation between players.

Players can minimize their walking time and improve overall efficiency by learning how to collaborate effectively.

Cramped Room. The Cramped Room layout presents a simplistic environment in which two players share an open room with a single pot at the top and a serving area at the bottom right. In this setup, players could score high even without extensive coordination.

Coordination Ring. The Coordination Ring layout is another ring-shaped kitchen, similar to the Counter Circuit. However, this layout is considerably smaller than Counter Circuit, with a close arrangement that makes it easier for players to complete soups. The ingredients, serving area, and plates are all in the bottom left corner, while the two pots are in the top right. As a result, this layout allows more easily achieving high rewards.

In summary, the Cramped Rm., Asymm. Adv., and Coord. Ring layouts are more conducive to achieving higher rewards when players cooperate with different partners. On the other hand, the Forced Coord., Counter Circ., and Asymm. Adv. layouts, due to their unique designs, provide more suitable testbeds for investigating and promoting cooperation among players.

6.2 Details on Human-AI Experiment Platform

We delve into the details of our Human-AI Experiment Platform, offering a comprehensive overview of its components and functionality. To better understand the system's user interface, layout, and functionality, a visual representation of the Human-AI Experiment Platform can be found in Appendix E.

As illustrated in Fig. 6, our human-AI experimental platform is composed of three primary elements: the front-end interface, the back-end server, and the database. The front-end interface is tasked with both rendering the game and recording real-time human keyboard input. Subsequently, at each frame, it engages in communication with the back-end server, transmitting the current game state and keyboard input as serialized data. On receiving this data, the back-end server initially processes it into an observation for the AI agent. Following this, it alters the game environment in response to the actions taken by both the AI agent and the human player. This newly altered game state is then conveyed back to the front-end interface for the human player's perusal. Simultaneously, the server captures the trajectories of both the human player and the AI agent during each game, storing this information for future analysis. Once the online experiments conclude, these captured trajectories are employed to conclude objective and subjective metrics.

Pre-experiment Pages. Before beginning the experiment, participants are presented with a page containing terms and conditions (e.g., experiment statement) and must decide if they agree to proceed. We provide the experimental statement used in our experiments as reported in Section 7, which includes information on Experimental Purpose, Procedure, Risks and Discomforts, Costs, and Confidentiality. If participants agree to the terms and conditions, they will be asked to complete a form with personal information, such as age, sex, level of skills in the game, etc. These data are used to facilitate a more in-depth analysis of the experiment results and will not be used for any purpose other than the experiment itself.

The instruction page then familiarizes participants with the mechanics of the game and the experiment process. Our example instruction page (Fig. 15) offers details on game